

Question:
At birth, the least developed part of the brain is the

Answer Choices:

- A. visual system
- B. thalamus
- C. frontal lobe
- D. brain stem
- E. parietal lobe
- F. cerebellum
- G. cortex
- H. basal ganglia
- I. hypothalamus
- J. limbic system

Reasoning + Answer:

<reasoning process here>

At birth , the brain undergoes rapid development , with certain regions developing earlier than others . The brain stem , which controls vital functions like breathing and heart rate , is one of the first structures to develop and is present at birth . The hypothalamus , thalamus , brain stem , basal ganglia , and limbic system are also present early , but the most developed structures in terms of neural circuitry and function at birth are the brain stem and subcortical regions .

However , the question asks for the *least developed* part of the brain at birth . The cortex , especially the prefrontal cortex , is the last part of the brain to mature . The frontal lobe , which is part of the cortex , is particularly underdeveloped at birth and continues to mature significantly during infancy and childhood . The cortex , especially the prefrontal regions , is responsible for higher cognitive functions and is the last structure to develop .

Therefore , the least developed part of the brain at birth is the cortex , particularly the frontal lobe , which is part of the cerebral cortex .

Among the options , the cortex (G) is the most accurate choice , as it is the most underdeveloped at birth . The frontal lobe (C) is a part of the cortex , and since the cortex as a whole develops last , it is the best answer .

However , since the frontal lobe is a specific part of the cortex and is especially underdeveloped , and since the cortex is the broad structure , the best answer is the cortex (G) , as it encompasses the development of the frontal lobe .

Thus , the least developed part of the brain at birth is the cortex .

<answer> G </answer>

